Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 PHYSICS

9478/02

26 August 2026

2 hours

Paper 2 Structured Questions

You must answer on the question paper.
No additional materials are needed.

INSTRUCTIONS

- Answer all questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and index number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do not use an erasable pen. Do not use correction fluid or tape.
- Do not write on any bar codes.
- You may use an approved calculator.

- **INFORMATION**
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiner's Use	
1	/10
2	/8
3	/14
4	/5
5	/10
6	/8
7	/20
s.f.	/−1
Total	/75

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
	$(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\mathrm{m\,F^{-1}})$
elementary charge	$e = 1.60 \times 10^{-19}\,\mathrm{C}$
Planck constant	$h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\mathrm{kg}$
rest mass of electron	$m_\mathrm{e} = 9.11 \times 10^{-31}\,\mathrm{kg}$
rest mass of proton	$m_\mathrm{p} = 1.67 \times 10^{-27}\,\mathrm{kg}$
molar gas constant	$R = 8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Avogadro constant	$N_\mathrm{A} = 6.02 \times 10^{23}\,\mathrm{mol^{-1}}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
acceleration of free fall	$g = 9.81\,\mathrm{m\,s^{-2}}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\mathrm{K} = T/^\circ\mathrm{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

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displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + ...$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + ...$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

1 An object of mass 0.050 kg is projected from the top of a 50 m tall building at an angle of 60° to the horizontal with an initial speed of 20 m s^{-1} as shown in Fig. 1.1.

A is a point in the trajectory that is at the same height as the object initially and *d* is the horizontal distance the object lands from the building.

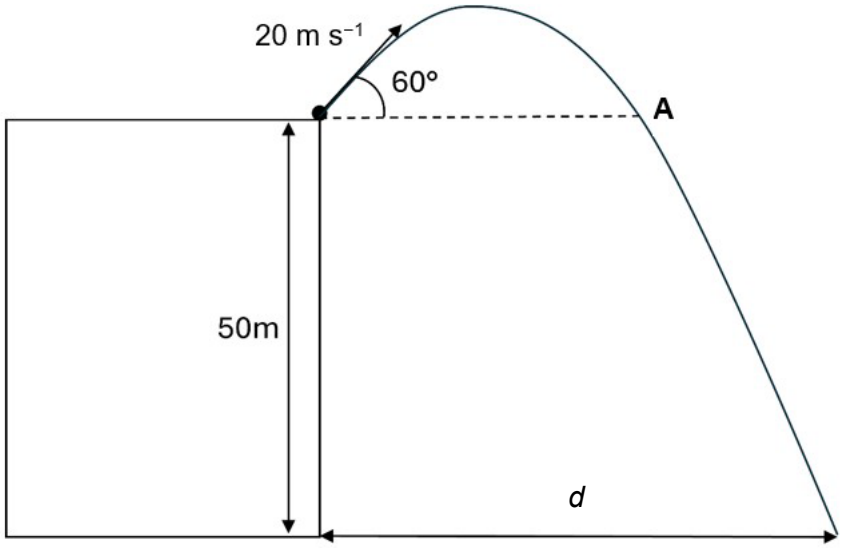


Fig. 1.1

(a) Assuming there is no resistance,

(i) Calculate the distance *d*.

taking the upward direction as positive,

vertical component of initial velocity, $u_y = 20\sin 60°$ (M1)

let t to be the time taken to reach the ground,

$s_y = u_y t + \frac{1}{2} a_y t^2$

$-50 = 20\sin 60° t - 4.905t^2$ (M1)

$t = 5.414$ s

$d = u_x t = (20\cos 60°)(5.414) = 54.1$ m (A1)

d = ... m [3]

(ii) Without calculation, state the speed of the object at **A** and explain how you arrived at your answer.

20 m s^{-1} [B1]

As the projectile rises to its maximum height, some of its kinetic energy is converted into gravitational potential energy. As it falls from the maximum height to point A (which is at the same height as the point of projection), the same amount of gravitational potential energy is converted back into kinetic energy.

Since mechanical energy is conserved (neglecting air resistance), the kinetic energy at point A is equal to the initial kinetic energy. Hence, the speed at A is the same as the initial speed. [B1]

OR

The projectile is at the same height initially and at point A, so its gravitational potential energy is the same at both positions.

As mechanical energy is conserved (neglecting air resistance), its kinetic energy must also be the same at these two positions. Therefore, its speed at A is equal to its initial speed. [B1]

..... [2]

(b) Experimentally, it is found that the same object now reaches a maximum height of 60.5 m above the ground in its trajectory.

(i) Show quantitatively that the air resistance is not negligible in this case.

neglecting the air resistance,
the maximum height s_y above the point of projection is given by:

$v_y^2 = u_y^2 + 2a_y s_y$

$0 = (20\sin 60°)^2 + 2(-9.81)s_y$

$s_y = 15.2905 \text{ m} = 15.3 \text{ m}$ (to 3 s.f.)

$\Rightarrow$ the maximum height above the building top $= 15.3$ m (M1)

factoring in air resistance,
the maximum height above the building top $= 10.5$ m

since 15.3 m > 10.5 m, (M1)
the air resistance is not negligible in the stated case

[2]

(ii) The speed of the object at its maximum height is 8.0 m s^{-1}.
Calculate the energy lost due to air resistance at maximum height.

initial kinetic energy, $KE_i = \frac{1}{2}mu^2 = 10.0$ J

kinetic energy at its maximum height, $KE_f = \frac{1}{2}mv^2 = 1.6$ J

gain in gravitational potential energy,
$\Delta GPE = mgs_y = (0.050)(9.81)(10.5) = 5.15$ J (M1)

By conservation of energy,
$KE_i = KE_f + \Delta GPE +$ energy lost due to air resistance
energy lost $= (KE_i - KE_f) - \Delta GPE = (10.0 - 1.6) - 5.15 = 3.25$ J (A1)

energy lost = ... J [2]

(c) Sketch the path of the object in Fig. 1.1 when air resistance is not negligible. [1]

[Total: 10]

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2 In a collision between two bodies A and B, the force that A exerts on B varies with time in the way shown in Fig. 2.1.

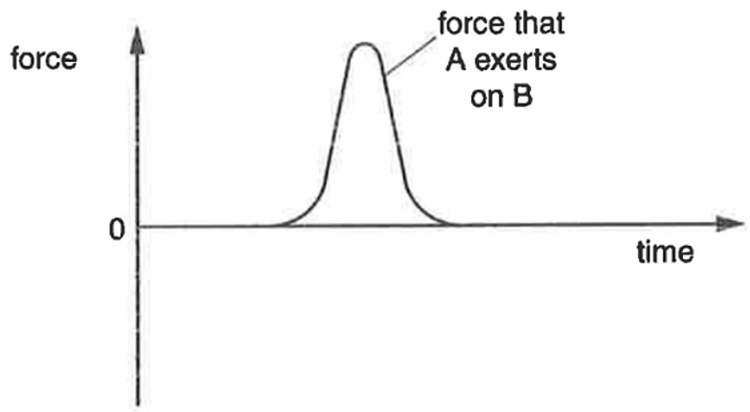
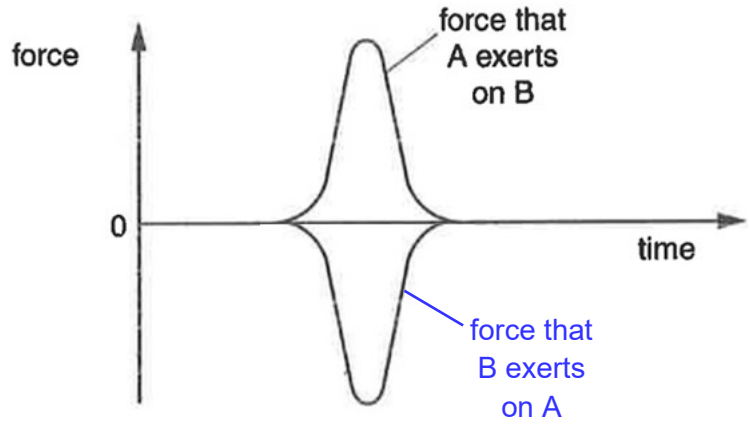


Fig. 2.1

(a) (i) Copy Fig. 2.1 and show in Fig. 2.2 a graph of the force that B exerts on A.



[B1]

Fig. 2.2 [1]

(ii) Explain your answer to **(i)**.

The forces are an action-reaction pair; therefore, they should always be equal in magnitude and opposite in direction. [B1]

……

……

…… [1]

(b) In a collision, when a truck of mass 12 000 kg runs into the front of a car of mass 1200 kg, a constant force of 72 000 N acts for 0.25 s.
Calculate the change in velocity of

(i) the car,

Since the truck runs into the front of the car (as stated), this describes a head-on collision.

impulse = $F \times \Delta t = 72000 \times 0.25 = 18000$ N s = 18000 kg m s^{-1} [B1]

Both the truck and the car experience equal and opposite impulses (Newton's Third Law) or change in momentum.

change in velocity of the car, $\Delta v = \Delta p / m_{car}$
$= (F\Delta t)/m_{car} = 18000/1200 = 15$ m s^{-1}, [B1]

in the direction of the truck's initial motion.

change in velocity = ... m s^{-1} [2]

(ii) the truck,

impulse = $F \times \Delta t = 72000 \times 0.25 = 18000$ N s = 18000 kg m s^{-1} [B1]

change in velocity of the truck, $\Delta v = \Delta p / m_{truck}$
$= (F\Delta t)/m_{truck} = 18000/12000 = 1.5$ m s^{-1}, [B1]

In the direction opposite to its own initial motion.

change in velocity = ... m s^{-1} [2]

(c) State and explain how the force on the driver could be reduced by safety features in a typical car.

Seat belts, airbags and crumple zones increase the time taken for the driver to come to rest. [B1]

For the same change in momentum, increasing the time of impact reduces the average force acting on the driver. [B1]

.....

..... [2]

[Total: 8]

3 An experiment is carried out, using a light intensity meter attached to a toy train, to measure the diffraction of laser light through a double slit, as shown in Fig. 3.1 below.

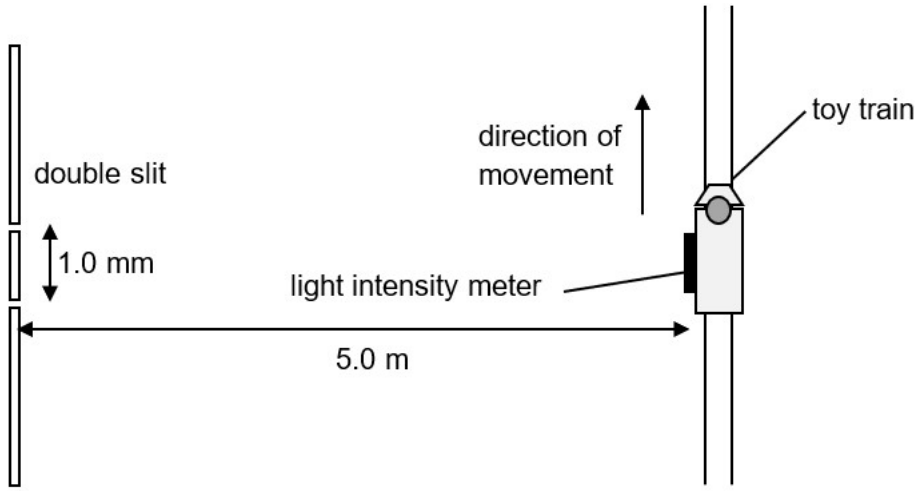


Fig. 3.1

Light from the laser is incident on the double slit. The light from the double slit is then detected by a light sensor attached to the train.

As the train moves along the track, the light intensity is recorded by the light intensity meter. The variation with time of the light intensity is shown in Fig. 3.2.

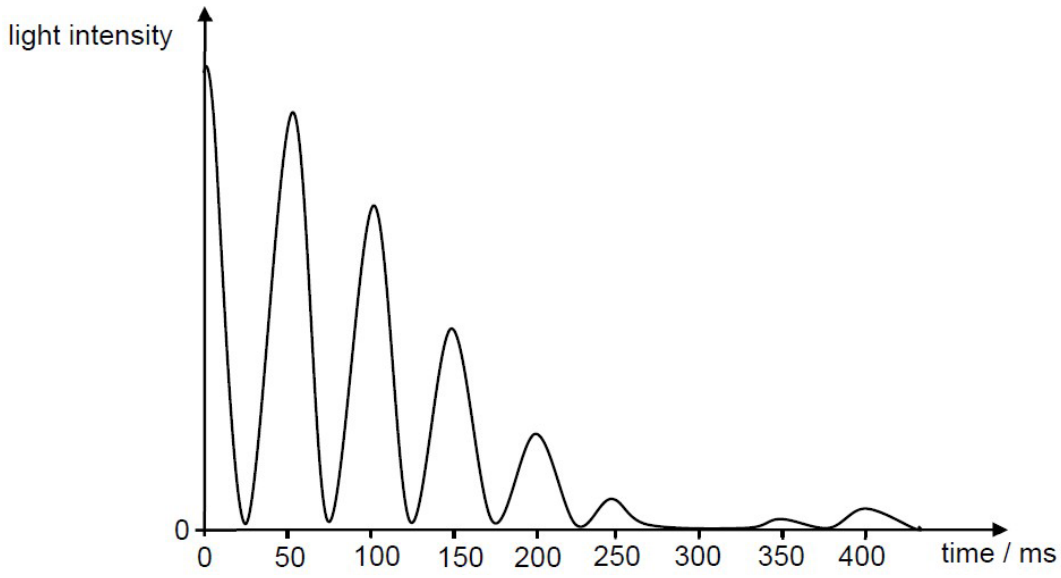


Fig. 3.2

(a) Explain, with reference to the paths travelled by the light emerging from each slit, why maxima and minima are observed.

Light from each slit travels a different distance to the sensor. When the path difference from each slit to the light intensity sensor is $n\lambda$ where n is an integer, the waves meet in phase and constructive interference occurs, light intensity reaches maximum value. [B1]

When the path difference from each slit to the light intensity sensor is $\left(n+\frac{1}{2}\right)\lambda$ where n is an integer, the waves meet antiphase and destructive interference occurs, light intensity reaches minimum value. [B1]

..... [2]

(b) **(i)** The speed of the train is 6.5 cm s^{-1}.
Use Fig. 3.2 to show that the fringe separation of the two slit interference pattern is 3.3×10^{-3} m.

fringe separation, $x = \left(\frac{200\times10^{-3}\ \text{s}}{4}\right)\times\left(6.5\times10^{-2}\ \text{m s}^{-1}\right) = 3.25\times10^{-3}\ \text{m}$ [B1]

(do not accept answer based on 50 ms for one fringe spacing directly)

[Students should measure over several fringe spacings and divide by the number of intervals to reduce the percentage uncertainty in the fringe separation.]

[1]

(ii) The separation of the slits is 1.0 mm and the track is 5.0 m from the slits.
Determine the wavelength of the laser light.

Since the slit-to-screen distance is much greater than the slit separation, i.e., $D \gg a$, we have $x = \frac{\lambda D}{a}$. Therefore,

$$\lambda = \frac{ax}{D} = \frac{\left(1.0\times10^{-3}\right)\left(3.25\times10^{-3}\right)}{5.0}$$
$$= 6.5\times10^{-7}\ \text{m}$$

wavelength = ... m [2]

(c) A maximum is detected by the light intensity meter at every 50 ms, explain why a minimum is detected at 300 ms in Fig. 3.2.



The double slit interference pattern is modulated / modified / enveloped by the single slit diffraction pattern.

OR There is an envelope because of (single) slit diffraction

At 300 ms the light sensor is at the first minimum of the single slit diffraction envelope where the light intensity is zero (due to destructive interference between wavelets from different parts of each slit), so no maximum interference fringe is observed at that position. [B1]

..... [1]

(d) Determine the width of one of the slits.

position of first minimum, $y = (300 \times 10^{-3}\ \text{s})(6.5 \times 10^{-2}\ \text{m s}^{-1}) = 0.0195\ \text{m}$

single-slit diffraction first minimum: $b \sin\theta = \lambda$

small angle approximation: $\sin\theta \approx \tan\theta = \frac{y}{D}$

$\therefore b\left(\frac{y}{D}\right) \approx \lambda$

$\Rightarrow b \approx \frac{\lambda D}{y} = \frac{(650 \times 10^{-9})(5.0)}{0.0195} = 1.7 \times 10^{-4}\ \text{m}$

width = ... m [2]

(e) The experiment is changed so that the toy car is now used to investigate the formation of a stationary sound wave as shown in Fig. 3.3 below.

The light sensor of the train is replaced with a sound sensor attached to a data logger. A loudspeaker emitting sound waves of constant amplitude and frequency is placed such that it faces a reflecting barrier. The train track is adjusted so that the train is moving away from the loudspeaker towards the barrier.

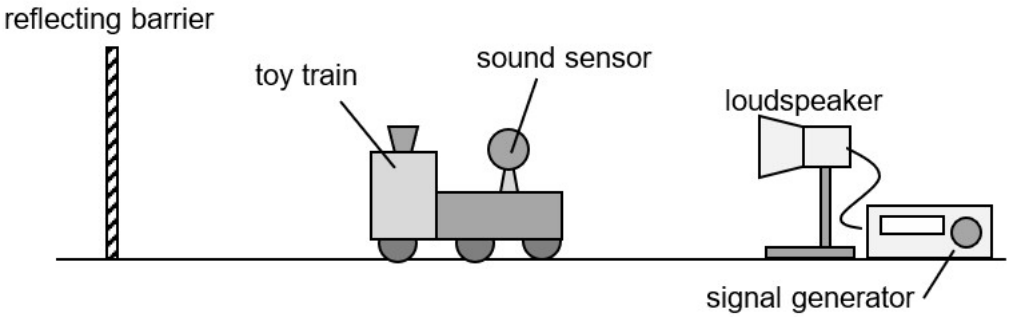


Fig. 3.3

The variation with distance of the sound intensity recorded by the sound sensor is shown in Fig. 3.4.

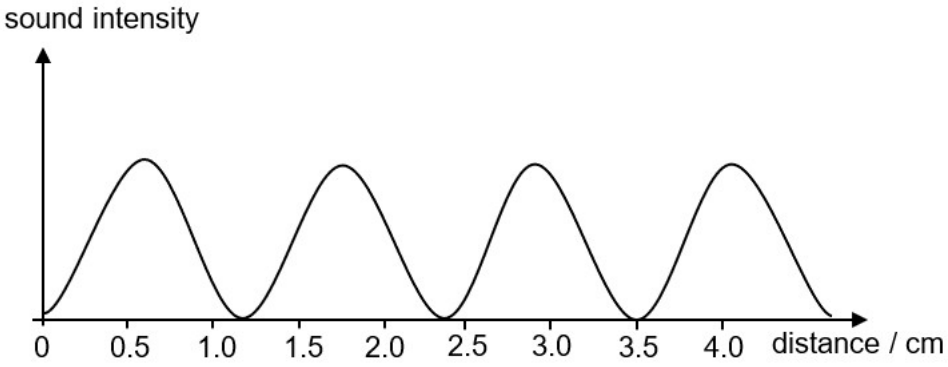


Fig. 3.4

(i) Explain how a stationary sound wave forms between the loudspeaker and reflecting barrier.

Sound wave from the loudspeaker is reflected by the barrier. [B1]

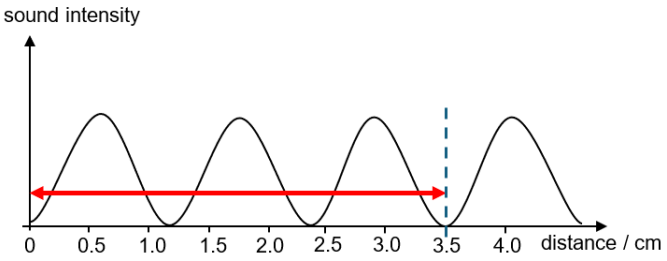
The reflected wave interferes / undergoes superposition with the incident wave. The resultant wave is a stationary wave. [B1]

…… [2]

(ii) Using Fig. 3.4, determine the wavelength of the stationary wave.

using separation between 4 consecutive nodes (1.5λ) as the distance from the graph, [B1]

wavelength, $\lambda = (3.5 \times 10^{-2}\ \text{m}) \div 1.5 = 0.0233\ \text{m}$ [A1]



wavelength = ... m [2]

(iii) Given that the frequency of the sound is 14 kHz.
Determine the speed of the sound wave.

speed, $v = f\lambda = (14000\ \text{Hz})(0.0233\ \text{m})$ [B1]

$= 326\ \text{m s}^{-1}$

$= 330\ \text{m s}^{-1}$ (to 2 sig. fig.) [A1]

speed = ... m s^{-1} [2]

[Total: 14]

4 A technician is designing a charging circuit for a compact camera flash unit as shown in Fig. 4.1. The circuit consists of a DC battery V of 12 V, a resistor R = 15 kΩ and an uncharged capacitor C of 220 μF, connected in series with a switch.

At time t = 0, the switch is closed.

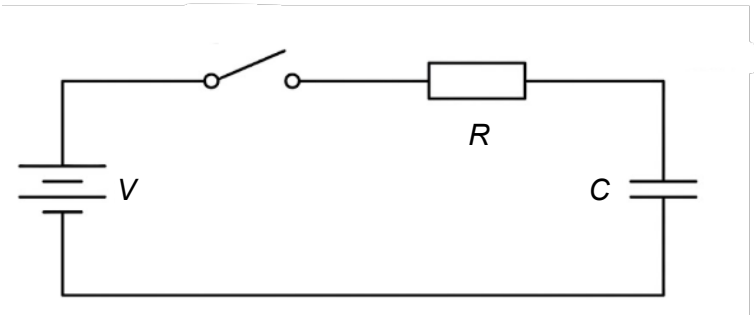


Fig. 4.1

(a) Calculate the time constant for the circuit.

The time constant of a series RC circuit is the product of the resistance and the capacitance:

$RC = 15 \times 10^3 \times 220 \times 10^{-6} = 3.3$ s [A1]

time constant = ... s [1]

(b) Calculate the percentage of maximum charge in the capacitor 6.6 s after the switch is closed.

$$Q = Q_0\left(1 - e^{-\frac{t}{RC}}\right) \Rightarrow \frac{Q}{Q_0} = \left(1 - e^{-\frac{6.6}{3.3}}\right) \quad \text{[B1]}$$

$$= 0.8647 \approx 87\% \quad \text{[A1]}$$

percentage = ... % [2]

(c) Hence or otherwise, determine the current flowing through the circuit exactly 6.6 s after the switch is closed.

$$Q \propto V_C \Rightarrow \frac{V_C}{V_0} = 0.8647 \quad \text{[B1]}$$

$$V_C = 0.8647(12) = 10.376 \text{ V}$$

$$V_R = 12 - V_C = 1.624 \text{ V}$$

$$I = \frac{V_R}{R} = \frac{1.624 \text{ V}}{15 \times 10^3\ \Omega} = 0.108 \text{ mA} \quad \text{[A1]}$$

alternatively:

charging current, $I = I_0 e^{-\frac{t}{RC}}$

$$I = \frac{V_0}{R}\left(e^{-\frac{6.6}{3.3}}\right) = 0.108 \text{ mA}$$

current = ... mA [2]

[Total: 5]

5 A circuit is set up as shown in Fig. 5.1. The circuit contains a battery of e.m.f. 12 V, and a fixed resistor r of resistance 6 Ω and length L_r = 0.50 m. A movable metal rod M, of length L_M = 0.60 m and resistance 12 Ω, is placed on a conducting track at a distance d = 4.0 cm from r. The conducting track is assumed to be smooth and has zero resistance.

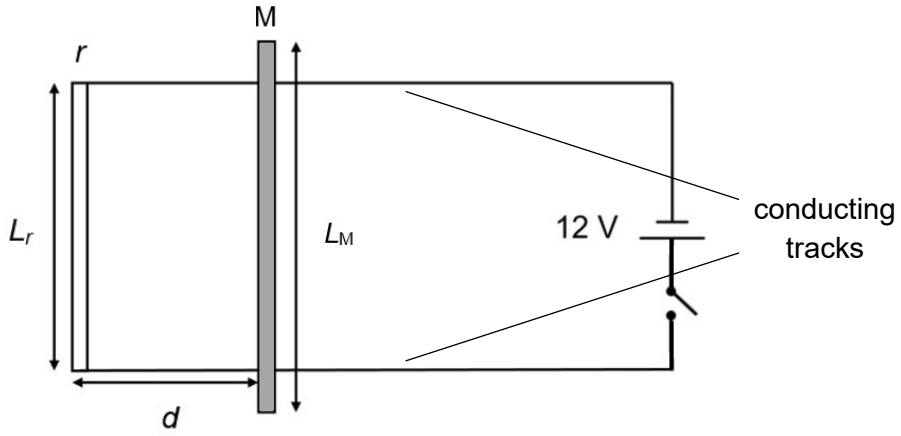


Fig. 5.1 (not to scale)

When the switch is now closed.

(a) Show that the magnetic flux density B at the metal rod M due to the current in resistor r is 10 μT.

the current passing through r, $I_r = \frac{12\text{ V}}{6\ \Omega} = 2.0\text{ A}$

$B = \frac{\mu_0 I_r}{2\pi d} = \frac{(4\pi \times 10^{-7}) \times 2.0}{2\pi(0.040)} = 1.0 \times 10^{-5}\text{ T} = 10\ \mu\text{T}$ (shown) [B1]

[1]

(b) Calculate the magnetic force experienced by M and state its direction.

$I_M = \frac{12\text{ V}}{(\frac{0.50}{0.60})(12\ \Omega)} = 1.2\text{ A}$

The magnetic force on M, acting over the current-carrying length of 0.50 m within the field of r:

$F = BI_M L = (1.0 \times 10^{-5})(1.2)(0.50)$ [M1]

$= 6.0 \times 10^{-6}$ N [A1]

Direction of F: leftwards, from Fleming's left-hand rule. [B1]

magnetic force = ... N

direction: ..... [3]

(c) M in **(b)** is held stationary and a magnetic field of flux density 1.0 mT is now directed out of page throughout the circuit in Fig. 5.1. M is now released.
Describe and explain the initial motion of M.

M moves and accelerates to the right. [B1]

The **applied magnetic field of 1.0 mT is much larger than 10 μT produced by r**. By Fleming's left-hand rule, the **applied field exerts a force to the right on M, which is larger than the 6.0 × 10^{-6} N force to the left due to the field of** *r*. Hence, the net force on M is to the right. [B1]

..... [2]

(d) A proton enters the magnetic field *B* at an angle to the field of less than 90°.
The proton follows the path shown in Fig. 5.2.

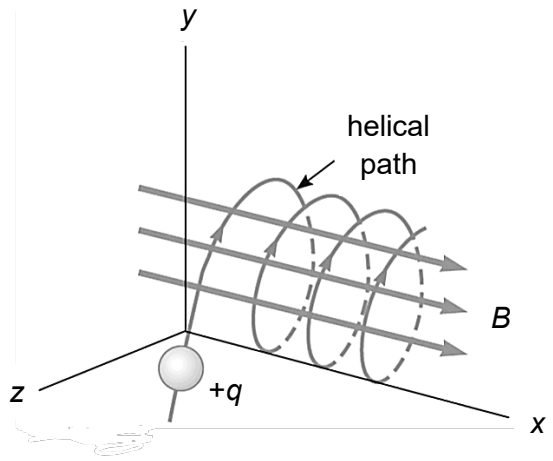


Fig. 5.2

(i) Explain why the proton follows this helical path.

Since the proton enters at an angle to *B*, its velocity has two components: **one perpendicular to *B* and one parallel to *B***. [B1]

The perpendicular component **experiences a magnetic force** ($F = Bqv\sin\theta$) that is always **perpendicular to its velocity**, **providing the centripetal force** for uniform circular motion. The parallel component experiences **no magnetic force** and so the proton continues to move along the direction of *B* at **constant speed**. The combination of these two motions produces a helical path. [B1]

.....

.....

..... [2]

(ii) Fig. 5.3 shows a magnetic bottle, where a non-uniform magnetic field is produced between two current carrying coils. A proton is initially directed perpendicularly into the page with speed *v* at the position shown.

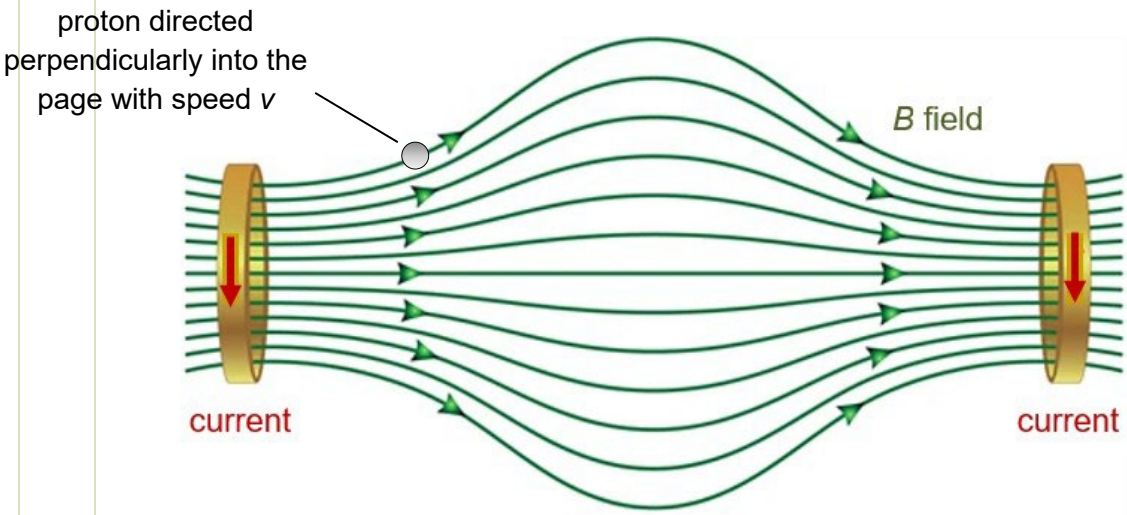


Fig. 5.3

Describe and explain the subsequent motion of the proton in the magnetic bottle, assuming the proton remains within the field region throughout.

The proton will experience a magnetic force perpendicular to the magnetic field based on Flemming's left hand rule. This **magnetic force, pointing towards the centre of the magnetic bottle**, produces velocity parallel and perpendicular to the magnetic field. **Past the centre** of the magnetic bottle, the magnetic force perpendicular to the field, **pointing towards the centre of the magnetic bottle**, opposes the velocity parallel to magnetic field, (reducing its magnitude and changes the direction of velocity eventually) [B1*]

This combined motion of a velocity parallel and perpendicular to the magnetic field in the non-uniform magnetic field produces a **spiral motion** from **one end of the magnetic bottle to the other** where the **radius of the motion increases towards the middle and decreases towards the ends**.[B1]

This process continues and the proton oscillate indefinitely between the two coils, while spiralling around the axis of the bottle.

[To take note] Throughout the motion, since the magnetic force is always perpendicular to the velocity, it does no work on the proton and the speed of the proton remains constant — only the distribution between the velocity components parallel and perpendicular to the B field such that the kinetic energy remains constant.

[Total: 10]

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6 (a) (i) Explain what is meant by the term *nuclear binding energy*.

- The binding energy of a nucleus is the minimum energy required to completely separate to infinity all the nucleons of a nucleus. [B1]
- The binding energy is the energy equivalent of the mass defect Δm of the nucleus: binding energy = $(\Delta m)\, c^2$ [B1]

.....[2]

(ii) A nucleus of strontium-90 has a mass of 89.907 *u*. It contains 38 protons. The mass of a neutron is 1.00867 *u* and the mass of a proton is 1.00783 *u*.

Calculate the nuclear binding energy per nucleon of strontium-90.

mass of 38 protons = 38.2974; mass of 52 neutrons = 52.4508; total 90.7484	
mass defect = 90.7484 – 89.907 = 0.841 *u*	[B1]
0.841 *u* × 1.66 × 10^{-27} kg × $(3.00 \times 10^8)^2$ = 1.26 × 10^{-10} J = 1.40 × 10^{-12} J per nucleon (3 s.f.)	[B1]

binding energy per nucleon = ... J [2]

(b) The nuclear fission of uranium-235 can be described by

$$^{1}_{0}\text{n} + {}^{235}_{92}\text{U} \longrightarrow {}^{144}_{56}\text{Ba} + {}^{89}_{36}\text{Kr} + 3{}^{1}_{0}\text{n}.$$

Table 6.1 shows the masses of the particles involved in this reaction.

Table 6.1

particle	mass / u
$^{1}_{0}\text{n}$	1.008665
$^{89}_{36}\text{Kr}$	88.917636
$^{144}_{56}\text{Ba}$	143.922953
$^{235}_{92}\text{U}$	235.043930

Determine the energy released by the fission of 15.0 kg of uranium-235.

mass defect, m = [235.043930 – (143.922953 + 88.917636 + (2×1.008665))] u = 0.186011 u	[B1]
$E = m c^2$ and m = (mass in u) × (1.66×10^{-27} kg u^{-1}) = [$0.186011 \times 1.66 \times 10^{-27}$] × ($3.00 \times 10^{8}$ m s^{-1})^2 = 2.78×10^{-11} J	[B1]
Number of atoms, $N = N_A \times \dfrac{\text{mass of sample}}{\text{molar mass}}$ or $\dfrac{\text{mass of sample}}{\text{mass of 235 nucleons}}$ $= 6.02 \times 10^{23}\ \text{mol}^{-1} \times \dfrac{15000\ \text{g}}{235\ \text{g mol}^{-1}}$ or $\dfrac{15.0\ \text{kg}}{235 \times 1.66 \times 10^{-27}\ \text{kg}}$ $= 3.84 \times 10^{25}$ or 3.85×10^{25}	[B1]
energy released = $(3.85 \times 10^{25})(2.78 \times 10^{-11}\ \text{J}) = 1.1 \times 10^{15}$ J	[B1]

energy = ... J [4]

[Total: 8]

7 Read the passage and answer the questions that follow.

Traditional wind turbines are limited by their height; the friction of the Earth's surface slows wind speeds at low altitudes, and the massive concrete foundations required for towers over 150 metres tall are both expensive and environmentally invasive. To overcome these limitations, engineers have developed **Airborne Wind Energy Systems (AWES)**.

In early 2026, the S2000 Stratosphere Airborne Wind Energy System (SAWES), developed in China, became the world's first megawatt-class "flying power plant" to successfully feed electricity into a national grid. The system utilizes a helium-filled, ring-shaped aerostat (a "balloon") anchored to a ground station by a high-strength tether. A simplified schematic of the S2000 system is shown in Fig. 7.1. The aerostat remains aloft through upthrust, the upward force exerted by the surrounding air on the balloon when it is immersed in the air. This allows the turbines to access the "jet effect" of the stratosphere—a region where winds are faster, more stable, and carry significantly more kinetic energy than those at ground level.

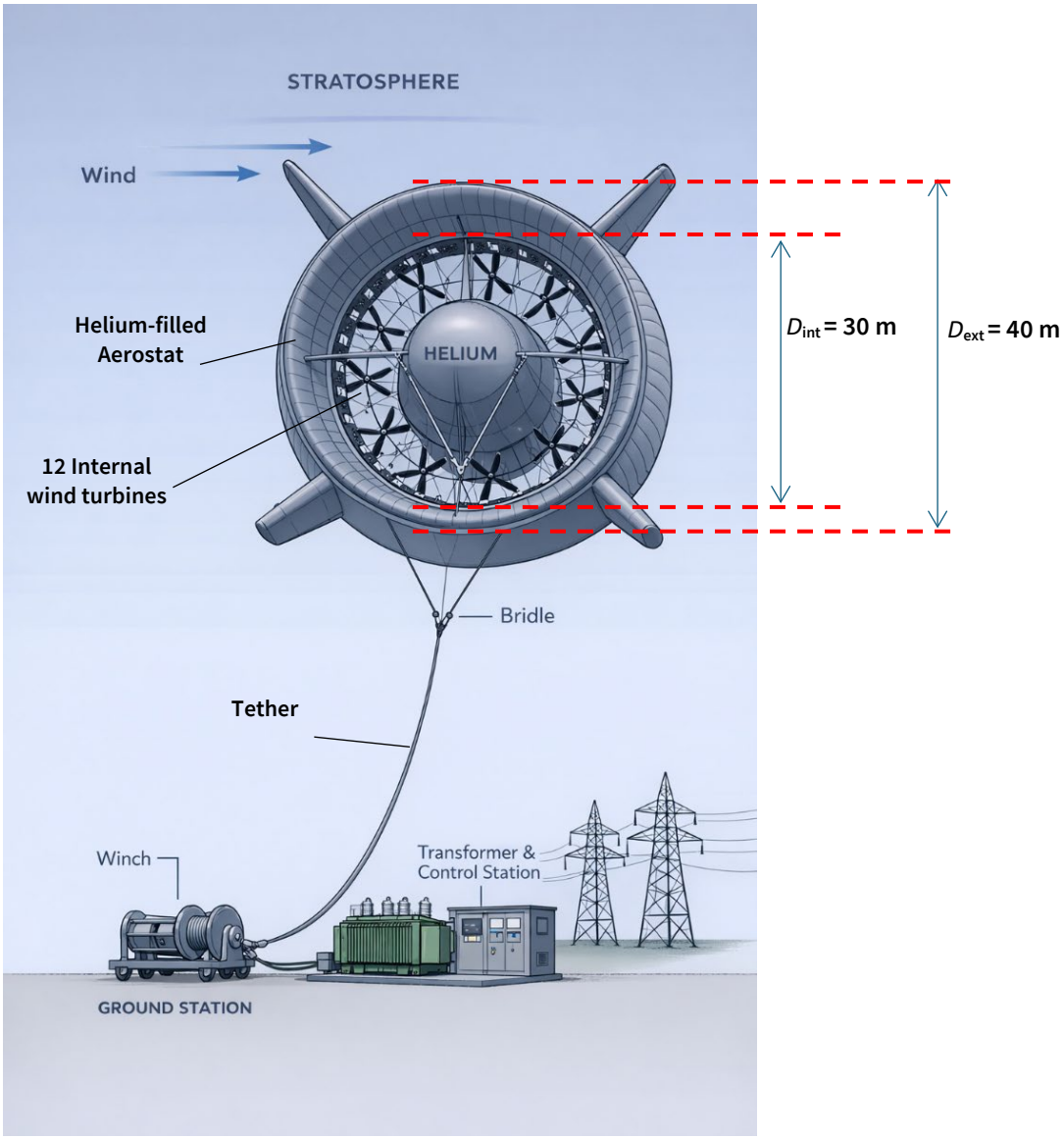


Fig. 7.1: Simplified Schematic of the S2000 System

The S2000 is linked to its ground-based winch station by a 2.5 km multipurpose conductive tether. This cable is a composite structure, featuring a high-tensile carbon-fibre core to withstand buoyancy and wind-drag forces, surrounding an aluminium-based conductive layer for power transmission. Onboard sensors relay real-time data through fibre-optic filaments within the tether, allowing a ground-based AI to adjust turbine speeds and aerostat pitch.

Table 7.1: S2000 SAWES System Specifications and Operating Parameters

Category	Parameter	Value (at h = 2000 m)
Atmospheric	Air density (ρ_{air})	1.007 kg m^{-3}
	Average Wind Speed (v)	15.0 m s^{-1}
Aerostat	External Diameter (D_{ext})	40 m
	Internal Diameter (D_{int})	30 m
	Volume of Helium (V_{He})	20,000 m^3
	Aerostat Structural Mass ($m_{structure}$)	5,000 kg
Generation	Total Turbine Swept Area ($A_{turbine}$)*	340 m^2
	Turbine Assembly Mass ($m_{turbine}$)*	2,400 kg
	Maximum Rated Power (P_{max})	3.0 MW
	Average Power Output from Initial Test	770 kW (over 30 min)
Tether	Total Length l_{tether}	2500 m
	Linear Mass	1.8 kg m^{-1}

*Note: Total of the 12 internal turbines.

Additional Constants and Parameters:

Aluminium resistivity: $\rho_{Al} = 2.8 \times 10^{-8}\ \Omega$ m

Helium density: $\rho_{He} =$ 0.18 kg m^{-3}

Power coefficient: $C_p = 0.40$ (maximum)

System efficiency: $\eta_{system} = 45\%$ (maximum)

Drag coefficient (ring aerostat): $C_d = 0.25$

Energy density of the stratospheric wind, which represents the kinetic energy per unit volume of moving air, helps engineers assess the energy resource available at different altitudes and locations. This, in turn, will help to determine the amount of power that can be harnessed from the wind through the turbine.

Real turbines cannot extract all available wind power. The power coefficient (C_p) represents the fraction of wind power actually converted to mechanical power. For example, a C_p of 0.40 means the turbine converts 40% of the wind's kinetic energy into mechanical rotation. System efficiency (η_{system}) accounts for subsequent losses during electrical conversion and transmission, reducing the final delivered power. This includes losses in the generator, power electronics, and the resistance of the 2.5 km tether. A system efficiency of 45% means only 45% of the mechanical power extracted by the turbines reaches the ground station as usable electricity.

(a)	**(i)**	Using data from Table 7.1, calculate the buoyant force (upthrust) $\rho_{air} V_{He} g$ acting on the helium-filled aerostat at its operating altitude.
		$\rho_{air} V_{He} g = 1.007 \times 20{,}000 \times 9.81 = 197.57 \times 10^3$ N [A1]
		buoyant force = ... N [1]
	(ii)	Calculate the total weight of Helium-filled aerostat.
		Weight of helium = 0.18 × 20,000 × 9.81 = 35,316 N Weight of aerostat = 5,000 × 9.81 = 49,050 N Weight of turbines = 2,400 × 9.81 = 23,544 N Total weight = 107,910 N [A1] = 108, 000 N
		total weight = ... N [2]
	(iii)	Using your answers from **(a)(i)** and **(a)(ii)**, determine the minimum tension in the tether.
		minimum tension = 197,570 − 107,910 = 89,660 N = 90,000 N [A1]
		minimum tension = ... N [1]

	(iv)	Calculate the weight of the tether.
		weight of tether = 2,500 × 1.8 × 9.81 = 44,145 = 44,000 N
		weight = ……. N [1]
	(v)	State whether the force exerted by the Winch on the tether will be greater, equal or smaller than your answer in **(a)(iii)**, explain your answer.
		The force exerted by the Winch on the tether will be smaller. [A1] Since the tether is in equilibrium, (considering the forces on the tether), the force on the tether upwards by the aerostat [answer in (a)(iii)] is equal to the weight of the tether and the force exerted by the Winch on the tether (downwards) [M1] ……..…. [2]
(b)	**(i)**	When wind flows around the ring-shaped aerostat, it experiences a drag force given by $\frac{1}{2}\rho v^2 C_d A$, where A is the frontal area presented to the wind. Assuming the frontal area is a circle with external diameter D_{ext}, calculate the drag force acting on the aerostat in the stratospheric wind conditions.
		Frontal area = $\frac{\pi D_{ext}^2}{4} = \frac{\pi(40^2)}{4} = 1256.6\ \text{m}^2$ [A1] Drag force, $F_D = \frac{1}{2}\rho v^2 C_d A = \frac{1}{2}(1.007)(15.0)^2(0.25)(1256.6) = 35590\ \text{N} \approx 36\ \text{kN}$ [A1] [Both the frontal area calculation and the substitution into the drag force formula must be shown explicitly. Award zero if working is absent even if the answer is correct.]
		drag force = ……. N [2]
	(ii)	Determine the magnitude of the change in tension in the tether, when drag force acts on the aerostat. Explain your answer.
		Magnitude of the change in tension in the tether = 36 kN [A1] Tension is a vector, a change in tension $\Delta \boldsymbol{T}$ will be vector subtraction given by $(\boldsymbol{T}_f - \boldsymbol{T}_i)$, which will be the opposite of the drag force. Therefore, magnitude of the change in tension $\lvert\Delta \boldsymbol{T}\rvert$, is 36 kN. [B1] Alternatively, since $\boldsymbol{T}_f = \boldsymbol{T}_i - \boldsymbol{D}$, we have $\boldsymbol{T}_f - \boldsymbol{T}_i = -\boldsymbol{D}$, which implies that the drag force D causes the change in tension in the tether.
		magnitude of change in tension = ……. N [2]

(c)	**(i)**	By considering the energy density of the stratospheric wind, show that the power that can be harnessed from the wind can be expressed as $\frac{1}{2}\rho_{air}A_{turbine}v^3$, where ρ_{air}, $A_{turbine}$ and v are quantities expressed in Table 7.1.
		energy density $= \left(\frac{1}{2}mv^2\right)_{air} \Big/ V_{air} = \frac{1}{2}\rho_{air}v^2$ [A1] $P = (\text{energy density})(\text{air volume change rate}) = \left(\frac{1}{2}\rho_{air}v^2\right)(A_{turbine}v) = \frac{1}{2}\rho_{air}A_{turbine}v^3$
		[2]
	(ii)	Determine the total power harnessed from the wind through the turbine.
		$P = \frac{1}{2}\rho_{air}A_{turbine}v^3 = \frac{1}{2}(1.007)(340)(15.0)^3$ $= 577.8\times10^3$ W [A1]
		total power = ... W [1]
	(iii)	Hence or otherwise, determine maximum electrical power delivered to ground.
		Mechanical power = $C_p \times P = 0.40\times577.8\times10^3 = 231.1\times10^3$ W [B1] Delivered power = $\eta_{system}\times231.1\times10^3 = 0.45\times231.1\times10^3 = 104.0\times10^3$ W [A1]
		power = ... W [2]
	(iv)	The average power output from the Initial test from Table 7.1 is 770 kW. State whether this is higher or lower than your calculated value in **(c)(iii)** and suggest a reason for the difference.
		Initial test shows 770 kW average (from Table 7.1), much higher than calculated value. [B1] Difference suggests variable wind conditions with periods of higher wind speeds during 30 min testing period. [B1] $\left\{P = (\text{energy density})(A_{turbine}v) = \frac{1}{2}\rho_{air}A_{turbine}v^3\right\}$ ..... [2]

(d)	Dimensions of the tether must be considered in order to carry a current of up to 300 A and heat losses through the tether only account for 0.48% of the maximum rated power $\left(P_{max}\right)$ for efficient transmission of power. Determine the cross-sectional area of the aluminium needed in the conductive layer of the tether that satisfies the above considerations.	
	$\frac{0.48}{100} = \frac{P_{loss}}{P_{max}} \Rightarrow P_{loss} = \frac{0.48}{100} \times \left(3.0 \times 10^6\right) = 14.4 \text{ kW}$ $P_{loss} = I^2 R \Rightarrow R = \frac{P_{loss}}{I^2} = \frac{14400}{300^2} = 0.16\ \Omega$ $R = \rho_{Al} \frac{l_{tether}}{A_{tether}}$ $\Rightarrow 0.16 = 2.8 \times 10^{-8} \left(\frac{2500}{A_{tether}}\right)$ $\therefore A_{tether} = 4.4 \times 10^{-4} \text{ m}^2$	
	cross-sectional area = …… m^2 [2]	
		[Total: 20]

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